



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE: PHYSICS
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mrs Govender
DATE	18 July 2023
MARKS	165
DURATION	2,5 hours

%

LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 20 pages and a separate data sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Additional space is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	10	Total
Mark											
Moderated Mark											
Question Total	14	24	10	23	9	24	19	20	12	10	165

Comment:

QUESTION 1**14 MARKS**

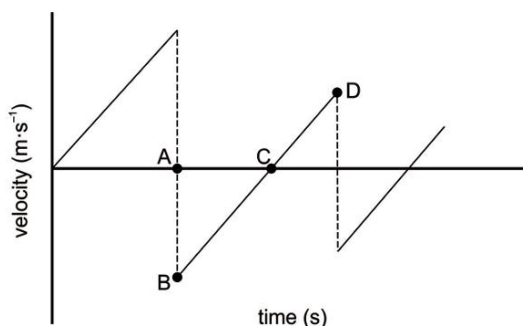
Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	☒	D
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Here the option C has been marked as an example.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D

- 1.1 The resultant of vectors **A** and **B** is **C**. **C** will have its greatest magnitude when: (2)
- A and B act at 180° to each other
 - A and B act at 90° to each other
 - A and B act at 45° to each other
 - A and B act at 0° to each other
- 1.2 A ball is projected vertically upwards from the ground at a speed of u . If air resistance is taken into consideration, the speed of at which the ball will hit the ground will be ...(2)
- Equal to 0
 - Equal to u
 - Smaller than u .
 - Bigger than u
- 1.3 A ball is dropped vertically from rest above a hard, horizontal surface. The motion for the bouncing ball is represented on the velocity vs time graph shown below. Air resistance is negligible. (2)



At which point labelled on the graph does the ball reach its maximum height after the first bounce?

- 1.4 A large crate of mass, m is placed on the flatbed of a truck but not tied down. As the truck accelerates forward with acceleration a , the crate remains at rest relative to the truck. What force relative to the ground causes the crate to accelerate? (2)



- A. The normal force
 B. The friction force
 C. The ma force exerted by the crate
 D. No force is required.
- 1.5 When four identical lamps P, Q, R and S are connected as shown in diagram 1, they have equal brightness.

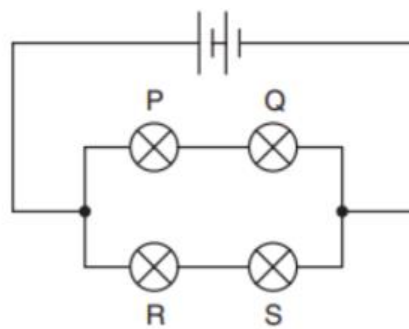


diagram 1

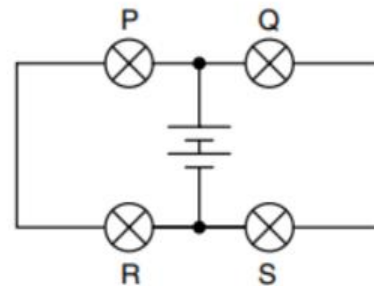


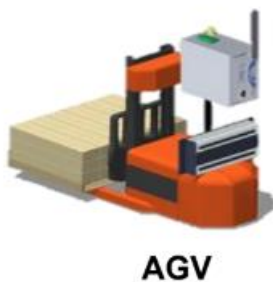
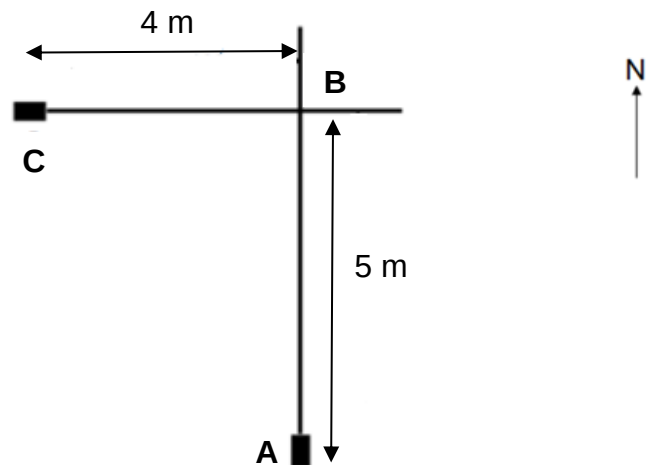
diagram 2

When the four lamps are connected as shown in diagram 2, which statement is correct? (2)

- A. The lamps do not light up.
 B. The lamps are dimmer than in diagram 1.
 C. The lamps have the same brightness as diagram 1.
 D. The lamps are brighter than diagram 1.
- 1.6 Newton's third law concerns the forces of interaction between two bodies. Which of the following statements relating to the third law is NOT correct? (2)
- A. The two forces must act on different bodies.
 B. The two forces are always in opposite directions.
 C. The two forces are at all times equal in magnitude.
 D. The two forces are equal and opposite so the bodies are in stationary.
- 1.7 A student weighing 500 N stands on a scale in an elevator (lift) moving down. When the scale reads 520 N, the elevator must be... (2)
- A. accelerating up
 B. increasing in speed
 C. moving at a constant speed
 D. accelerating down

QUESTION 2 KINEMATICS – HORIZONTAL MOTION**24 MARKS**

An automated guided vehicle (or **AGV**) shown in the diagram below uses magnetic tracks to move materials around a factory floor.

**AGV**

The vehicle begins moving along a track from **A** towards junction **B** as shown in the diagram below. The vehicle accelerates uniformly from rest at **A**, reaching a velocity of $0,5 \text{ m}\cdot\text{s}^{-1}$ at junction **B**, which is 5 m north of **A**. It then continues at this constant speed from **B** to **C** which is 4 m west of **B**.

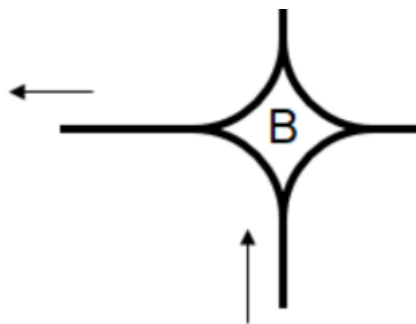
2.1 Define the term *displacement*. (1)

2.2 Calculate the magnitude of the acceleration of the vehicle as it moves from **A** to **B**. (3)

2.3 Calculate the total time it takes for the vehicle to travel from **A** to **C**. (4)

2.4 Calculate the average speed of the vehicle during the trip from **A** to **C**. (2)

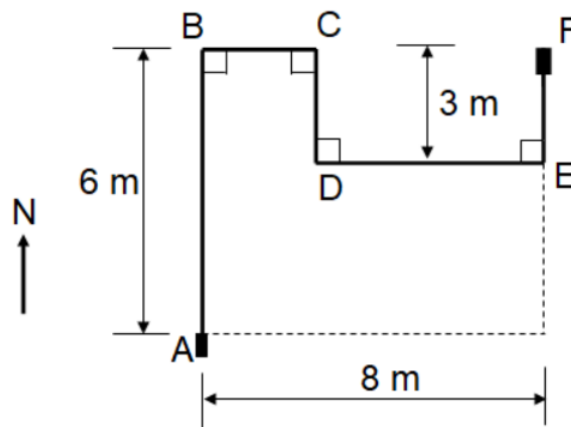
2.5 Consider junction **B**. A much closer view of junction **B** is shown below.



The vehicle travels at constant speed throughout its motion at junction **B**.

Did the vehicle accelerate as it turned left at junction **B**? Explain your answer. (2)

- 2.6 A vehicle moves from point A to F along path ABCDEF as shown in the diagram below. Point B is due north of A. The lengths of the various sections of track are shown in the diagram. The total time of the trip from point A to F is 10 minutes.

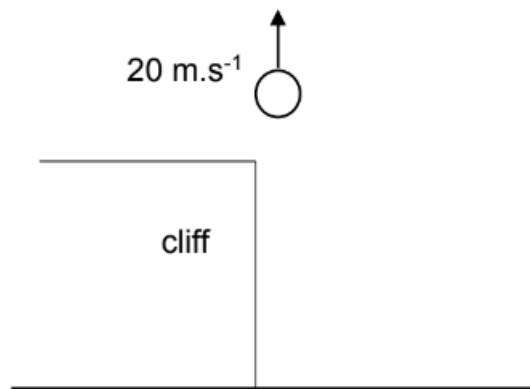


- 2.6.1 Calculate the total distance travelled by the vehicle from **A** to **F**. (3)
- 2.6.2 Determine the resultant displacement of the vehicle during the trip from point **A** to **F** as shown in the diagram. Give the direction as a bearing. (3)
- 2.6.3 Calculate the magnitude of the average velocity of the vehicle during the trip from point **A** to **F**. (2)
- 2.7 A golf ball travelling rectilinearly at a velocity of **Y** slows uniformly to come to rest after a distance **S**. At what initial velocity, **in terms of Y**, must it be struck to come to rest in **double the distance**, assuming that it experiences the same acceleration **a**? (4)

QUESTION 3 KINEMATICS – VERTICAL MOTION**10 MARKS**

A boy, standing on the edge of a cliff, projects a ball vertically upwards at 20 m.s^{-1} . On its return the ball passes the boy and reaches the ground at the bottom of the cliff 6 s after it is projected. Ignore the effects of air resistance. Assume that the point of release is at the edge of the cliff.

Take the upward direction as positive.



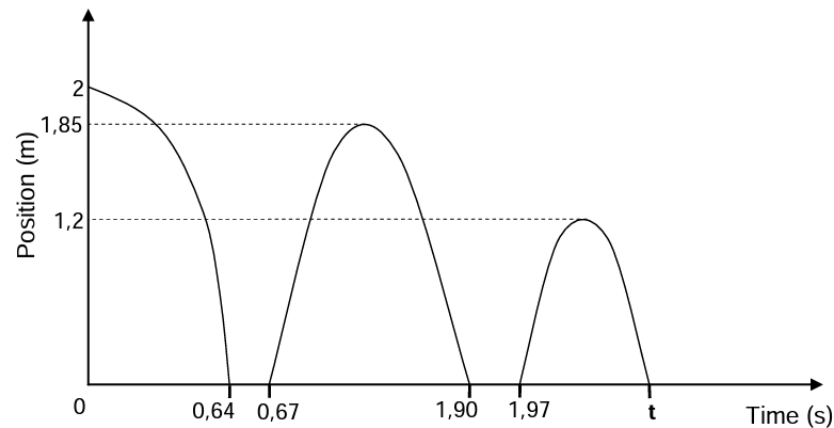
3.1 Define the term *velocity*. (2)

3.2 Calculate the time taken for the ball to pass the boy on its way downwards. (3)

3.3 Calculate the height of the ball, above the ground 2,8 s after it is released. (5)

QUESTION 4 KINEMATICS – VERTICAL MOTION**23 MARKS**

- 4.1 A small golf ball is dropped from a height of 2 m and bounces a few times after landing on a cement floor. Ignore air friction. The position-time graph below, not drawn to scale, represents the motion of the ball.



- 4.1.1 Explain what is meant by the term free fall. (2)

- 4.1.2 Describe the motion of the ball between 0s and 0,64s. (2)

- 4.1.3 Calculate the time that the golf ball is in contact with the floor before the first bounce. (1)

- 4.1.4 Calculate the time it takes the golf ball to reach its maximum height after the first bounce. (2)

4.1.5 Calculate the speed at which the golf ball leaves the floor at the first bounce.(3)

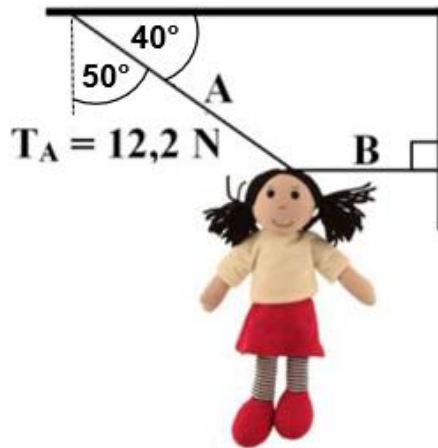
4.1.6 Calculate time t indicated on the graph. (5)

4.1.7 **Sketch** a velocity-time graph for the motion of the golf ball from time 0 s – 1,97 s. Clearly indicate the times of on the graph (no other values are required).(5)

4.1.8 Is energy lost during the bounce? Circle **YES** OR **NO**. Give a reason for the answer using information in the graph. (3)

QUESTION 5 NEWTON'S LAWS AND VECTORS**9 MARKS**

A doll is suspended by means of two light inextensible strings from the ceiling in the corner of the room as shown. String A makes an angle of 40° with the ceiling and string B makes an angle of 90° with the wall. The tension in string A is $12,2 \text{ N}$.



5.1 State *Newton's first law of motion*.

(2)

5.2 Draw a triangle of forces (vector addition diagram) for the forces acting on the doll. (3)

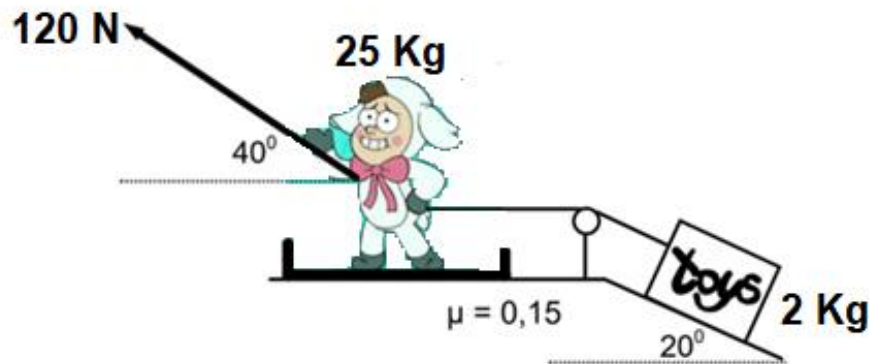
5.3 Hence, determine the mass of the doll.

(4)

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QUESTION 6 NEWTON'S LAWS**26 MARKS**

A cartoon character is standing on a little crate, dragging his toy box behind him using a rope over a frictionless pulley. Mr Ntombela's son decided to drag the crate across the floor towards the cricket centre. The kinetic coefficient of friction for both crate and toybox on the surface is 0,15.



6.1 Define the term *frictional force due to surface*. (2)

6.2 Draw a fully-labelled free body diagram of all the forces acting on the crate. (5)

6.3 Prove that the kinetic friction that the toy box is 2,76 N. (2)

6.4 Calculate the normal force experienced by the cartoon and the crate if they have a combined mass of 25 kg. (3)

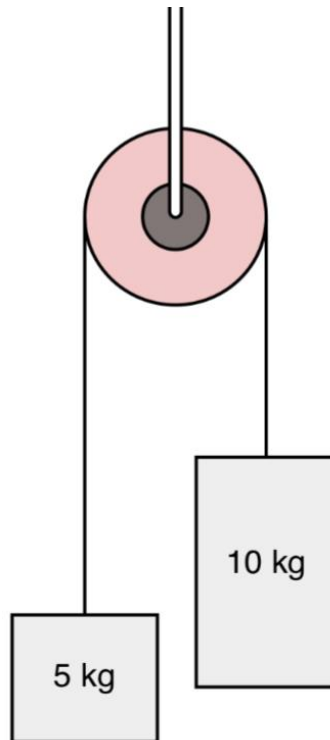
6.5 Calculate the magnitude of the kinetic frictional force that the cartoon and the crate experience as they are being pulled to the left by Mr Ntombela's son. (3)

6.6 Prove by calculation that the force exerted by Mr Ntombela's son is sufficient to accelerate the system towards the cricket centre at a rate greater than 2 m.s^{-2} . (6)

6.7 Luke is convinced that if Mr Ntombela's son exerts the same force at an angle of 0° his applied force on the crate would be greater and the frictional force that the crate experiences would be less, and therefore the system would experience a greater acceleration. Critically analyse his reasoning. (3)

QUESTION 7 NEWTON'S LAWS**19 MARKS**

- 7.1 Two masses are attached to a light inextensible cable which passes over a smooth pulley. The one mass is 10 kg and the other is 5 kg as shown in the diagram. The system is released from rest.



- 7.1.1 Calculate the magnitude of the acceleration of the system and the magnitude of the tension in the cable. (5)

- 7.1.2 State *Newton's third law of motion*. (2)

- 7.1.3 Give the name and direction reaction force to the weight of the 5 kg mass piece. (2)

- 7.2 Consider a brick on the inclined plane. The angle θ is increased gradually until the brick begins to slide.



- 7.2.1 Define the term *normal force*. (2)

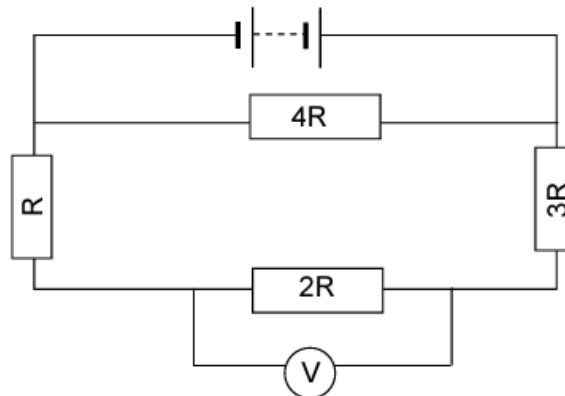
- 7.2.2 Write an expression in terms of θ for the normal force acting on the brick. (2)

- 7.2.3 Write an expression in terms of θ for the maximum static friction force ($F_{fs}^{\max}$) acting on the brick before it begins to slide. (2)

- 7.2.4 The 1,3 kg block is now placed onto a different rough surface with a coefficient of static friction of 0,48, inclined at an angle θ to the horizontal. The angle θ is now increased gradually, until the block starts to slide. Calculate the angle θ when the block is just about to slide down the slope. Show all your working out (4)

QUESTION 8 ELECTRIC CIRCUITS**20 MARKS**

Study the diagram and answer the questions that follow. The internal resistance of the battery and any resistance in the wires can be ignored.



8.1 State *Ohm's Law*. (2)

8.2 Calculate the value of resistor R if the total resistance of the circuit is $4,8 \, \Omega$ (3)

8.3 Define the term *potential difference*. (2)

8.4 Calculate the reading on the voltmeter if the current through the $4R$ resistor is $1,8 \, \text{A}$. (5)

8.5 Define the term *electrical work*. (2)

8.6 Calculate the amount of energy dissipated by the **4R** resistor in 2 minutes. (3)

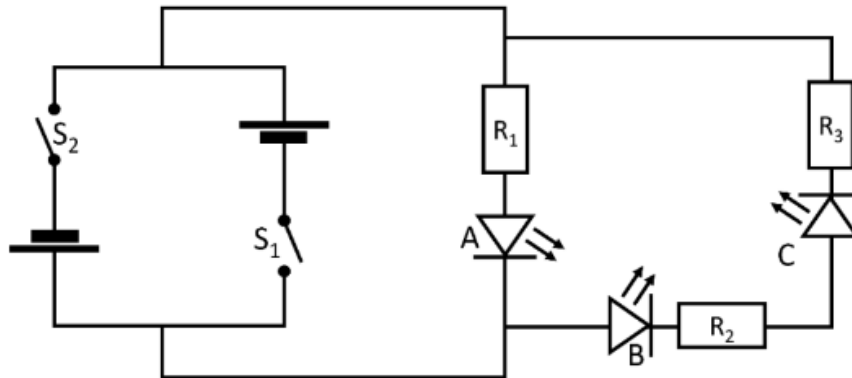
The **4R** resistor is replaced with an ammeter.

8.7 How will the reading on the voltmeter be influenced? Circle only INCREASE, DECREASE or STAY THE SAME. (1)

8.8 Explain the answer to QUESTION 8.7. (2)

QUESTION 9 ELECTRIC CIRCUITS**12 MARKS**

The following circuit is set up using three light emitting diodes (LEDs); A, B and C to illuminate a small ornament. The amount of the light produced is determined by opening or closing switches S_1 and S_2 . Resistors: R_1 , R_2 and R_3 all have an equal resistance.



9.1 Define *diode*. (2)

The LED A has a knee voltage of 2,5 V.

9.2 Briefly explain what a knee voltage is. (2)

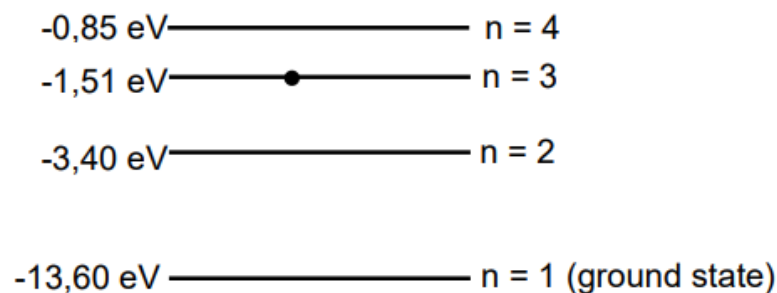
9.3 Which switch should be closed to produce the brightest possible light?
Circle **S_1** or **S_2** . (1)

9.4 Explain your answer to QUESTION 9.3 (2)

- 9.5 If the emf of both cells are 9 V what would the resistances of R_3 and R_2 need to be to ensure the potential difference over the LEDs B and C are equal to the knee voltage while allowing a current of 10 mA to flow through each LED? (5)

QUESTION 10 ELECTROMAGNETIC SPECTRUM**10 MARKS**

The energy levels for a hydrogen atom are represented on the diagram below.



An excited electron is in the 3rd energy level as indicated.

- 10.1 How many unique frequencies will be in the emission spectrum as the excited electron returns to the ground state? (2)
- 10.2 Calculate the frequency of the photon emitted when the electron transition is from the $n = 2$ level to the ground state. (4)

- 10.3 Hydrogen gas in a discharge tube emits light of wavelength 655 nm. Calculate the energy (in eV) that corresponds to this wavelength of light. (4)

[illegible]
